



The Impact of Climate Change on Mountain Weather

Course: PGE.4 Data Specialization

Module: 4DVST

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1. Introduction

- **Background:** As part of this project, we were tasked with analyzing one or more data sources from Météo-France's Open Data using GeoJSON for map visualization.
- **Research question:** Does climate change affect weather conditions in the mountains ?
- **Data sources:**
 - Open Météo France
 - Climatic change (monthly)
 - Mountain weather (daily)

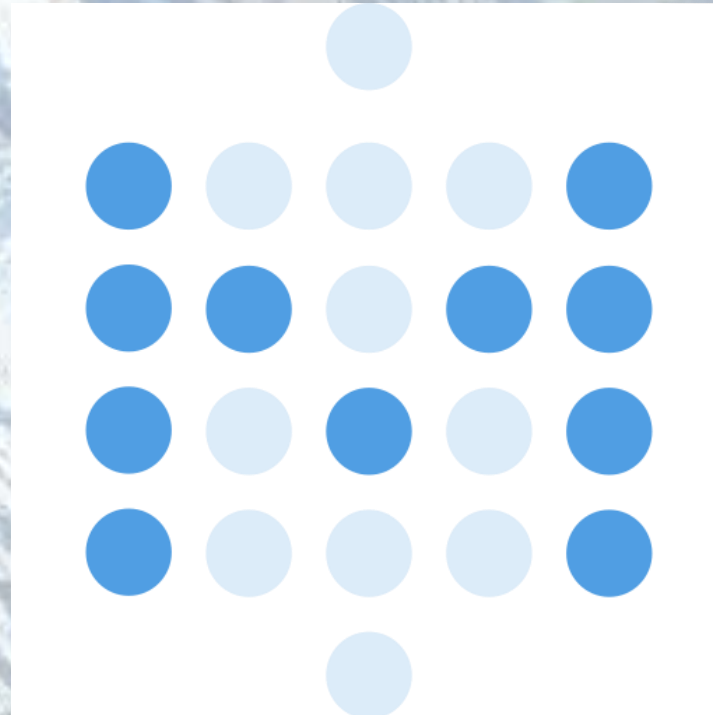


- French departments: <https://france-geojson.gregoiredavid.fr/repo/departements.geojson>
- French regions: <https://france-geojson.gregoiredavid.fr/repo/regions.geojson>

2. Technologies Used



dbt



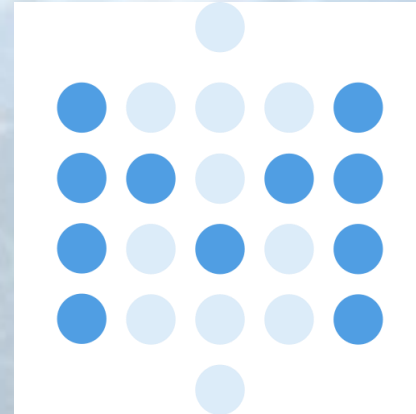
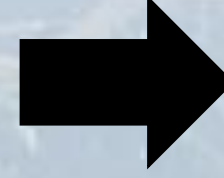
3. Overall Architecture



MENSUEL
(Climatic change)

NIVO
(Mountain weather)

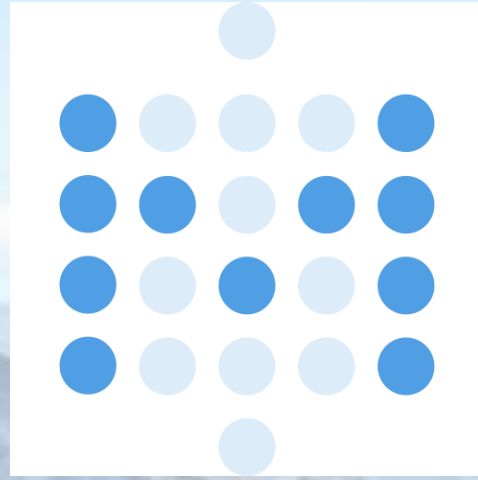
GeoJSON
(French regions and
departments)



4. Work Methodology

Key steps:

1. **CSV/GeoJSON extraction:** loading raw data into SQL
2. **Data cleaning and transformation:** cleaning and transforming data with DBT
3. **Analytical models:** creating analytical models for dashboards
4. **Visualizations and dashboards:** creating visualizations and dashboards with Metabase based on the analytical model



5. Demonstration

6. Conclusion

Current status:

- Two dashboards: one showing historical weather data and the other illustrating the impact of climate change on that weather
- A functional ELT process that can be deployed to production

Areas for improvement:

- More precise KPIs
- Improve query performance by creating indexes in SQL